

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE			
	Supplementary Examination – Summer 2022			
	Course: B. Tech.	Branch: E&TC	Semester: VI	
	Subject Code & Name: BTETC603 Digital Image Processing			
	Max Marks: 60	Date:	Duration: 3 Hr.	
	Instructions to the Students: 1. All the questions are compulsory. 2. Use of non-programmable scientific calculators is allowed. 3. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.		12	
A)	What are the fundamental steps in digital image processing?	(1)	6	
B)	Explain different types of neighbourhoods in image processing.	(2)	6	
C)	State Properties of 2D convolution and 2D Fourier Transforms.	(3)	6	
Q.2	Solve Any Two of the following.		12	
A)	Write a short note on image perception.	(2)	6	
B)	Draw and explain Structure of the Human Eye.	(1)	6	
C)	Define: 1. Brightness Sensitivity 2. Wavelength Sensitivity 3.Spatial Frequency Sensitivity.	(3)	6	
Q. 3	Solve Any One of the following.		12	
A)	What is image sampling and its types? Explain any 2 types of sampling in detail.	(2,3)	12	
B)	Explain quantization process in digital image processing in detail. What is Lloyd-Max quantiser?	(3,4)	12	
Q.4	Solve Any Two of the following.		12	
A)	Explain 2D DFT and DCT with its properties	(2)	6	
B)	What is smoothing and edge detection for 2D image?	(4)	6	
C)	Write a short note on WHT and KLT.	(2)	6	
Q. 5	Solve Any Two of the following.		12	
A)	Explain following Image enhancement techniques. i. Image negative ii. Log transformations iii. Power law transformations	(3,4)	6	
B)	How to reduce image noise and blurring in image restoration.	(4)	6	
C)	Write a short note on histogram processing and equalization.	(2,3)	6	

	*** End ***	
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The grid and the borders of the table will be hidden before final printing.